

Aditya Kulkarni

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| Raleigh, NC (Open to Relocation)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production full-stack experience in Python, React, and cloud-native AWS systems. At Isomer AI, shipped 10+ backend features in production -- 394 commits, 7 deployments, agile sprint delivery -- with real-time event pipelines, RBAC, and document processing. Built and test-driven Java and TypeScript APIs with Testcontainers concurrency validation and OpenTelemetry + Prometheus observability. Designed fault-tolerant distributed systems (Temporal.io + Kafka) handling 2,000 async requests/second. Proficient in Python, FastAPI patterns, React, AWS, Docker, and PostgreSQL; experienced in TDD, code reviews, and incident response with Sentry.

CORE COMPETENCIES

Python / FastAPI

React / Full-Stack

AWS Cloud Infrastructure

TDD / Clean Code Architecture

Scalable & Fault-Tolerant Systems

Observability & Incident Response

Agile / Sprint Delivery

Docker / CI/CD

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer -- Chicago, IL (Remote)

- Shipped **10+ production features in agile sprints** across a full-stack TypeScript/Node.js/PostgreSQL/Redis platform -- RBAC, real-time event pipelines, document processing (PDF, email via Microsoft Graph API, S3); Docker, GitHub Actions CI/CD, Sentry incident monitoring -- 394 commits, 7 deployments, PR reviews across engineering team
- Designed and implemented **Temporal.io workflow orchestration** and multi-provider LLM agent tooling (Anthropic Claude, OpenAI GPT-4o, Google Gemini) with retry/fallback logic; system handles production insurance claim workflows where failures have direct business impact

NC State University Apr 2025 - Present

Graduate Research Assistant -- Raleigh, NC

- Designed a cloud-based genomic data platform on Google Cloud Run serving 500GB datasets to 100+ researchers; collaborated on ML architecture with hyperparameter tuning, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern -- Mumbai, India

- Built 5+ **Python AI workflows** including health claim automation (OCR, NLP, ML); improved claim accuracy by 25%; automated deployments on **AWS (EC2, S3)** and GCP with Grafana monitoring and alerting dashboards

Capgemini Jun 2023 - Aug 2023

AI/ML Intern -- Mumbai, India

- Led end-to-end SDLC for a distributed data pipeline (Hadoop + Spark batch, Kafka streaming); reduced ticket-processing latency by 20% and processing time by 15%

PROJECTS

Inventory Reservation Service TDD + Java 21

Transactional reservation API built test-first: PostgreSQL row-level locking (SELECT FOR UPDATE) prevents overselling; idempotent endpoints with RFC 9457 error responses. Testcontainers TDD concurrency proof -- two concurrent reservations against quantity 1 produce exactly one success and one 409 Conflict. Flyway migrations, Spring Security (RBAC), multi-stage Docker build, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Security, Testcontainers, Docker, GitHub Actions

PulseRank -- Real-Time Leaderboard & Activity Feed Observability

Event-driven platform: MySQL transactional writes (idempotency key, outbox pattern), MongoDB feed projections (TTL expiry), Redis sorted sets for sub-millisecond reads. Live updates via Server-Sent Events. k6 load benchmarks, **OpenTelemetry + Prometheus** observability, Testcontainers multi-store consistency tests, GitHub Actions CI/CD.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus, k6

[AlgoPulse -- Fault-Tolerant AI Orchestration Platform](#) Distributed Systems

Identified API-polling bottleneck under load; re-architected to fault-tolerant event-driven model. Temporal.io manages durable workflow state with automatic retry; Kafka handles worker coordination. Result: 2,000 async inference requests/second with zero state loss under failure scenarios.
TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

[LLM Eval Harness](#) Python

Vendor-neutral Python benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 2,000-example frozen test set. Metrics: ExactMatch, TokenF1, hallucination rate, p50/p95 latency, cost/call with paired bootstrap CIs and McNemar testing.
Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

EDUCATION

North Carolina State University -- MS Computer Science	Aug 2024 - May 2026
Courses: Algorithms, Operating Systems, Software Engineering, Artificial Intelligence, Data Science, Object-Oriented Development	
Dwarkadas J. Sanghvi College of Engineering -- BTech Computer Science	Aug 2020 - May 2024
CGPA: 9.32/10 · Courses: DS & Algorithms, OS, Machine Learning, Big Data Infrastructure, Software Engineering	

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C, R
Backend & Frameworks: FastAPI, Node.js, NestJS, Spring Boot, REST APIs, Temporal.io, Kafka, Spark, Hadoop
Frontend: React, Next.js, Tailwind CSS, Server-Sent Events
Cloud & DevOps: AWS (EC2, S3), GCP (Cloud Run), Docker, Kubernetes, GitHub Actions, Terraform, OpenTelemetry, Prometheus, Grafana, Sentry
Databases: PostgreSQL, MySQL, MongoDB, Redis
Testing: TDD, Testcontainers, k6, Jest, pytest